

CHAPTER 7

Wireshark

WHAT YOU WILL LEARN IN THIS CHAPTER:

- Wireshark
- OSI Model
- Capture
- Filters and Colors
- Inspection

Wireshark

My first real experience using Wireshark was in a forensics class with Sherri Davidoff, CEO of LMG Security. Sherri walked us through many tools to investigate a case study where money had been stolen. Wireshark was the tool we kept returning to time and time again to prove what had been planned and executed, and eventually we were able to prove who the threat actors were.

Wireshark is a tool that every network or security administrator should know. It is an open-source tool used for capturing network traffic and analyzing packets at an extremely granular level. Sometimes Wireshark is called a *network analyzer* or a *sniffer*. Packet capturing can tell you about transmit time, source, destination, and protocol type. This can be critical information for evaluating events that are happening or troubleshooting devices across your network. It can also help a security analyst determine whether network traffic is a malicious attack, what type of attack, the IP addresses that were targeted, and where the attack originated from. As a result, you will be able to create rules on a firewall to block the IP addresses where the malicious traffic originated.

Wireshark shows packet details captured from different network media, breaking down the Open Systems Interconnection (OSI) model into the data link, network, transport, and application layers. At the bottom of the workspace, you have an option to open the hexadecimal with corresponding ASCII values on the right.

Wireshark is a powerful tool and technically can be used for eavesdropping. When you plan to use this in a business environment, you will want to get written permission to use it and make sure your organization has a clearly defined security privacy policy that specifies the rights of individuals using the network. Stories abound of network administrators capturing usernames, passwords, email addresses, and other sensitive user data. Wireshark is legal to use, but it can become illegal if you attempt to monitor a network that you do not have explicit authorization to monitor.

Determining the resources that Wireshark needs depends on the size of the .pcap file you are examining. If you have a busy network, then the files will be large. Wireshark can run on Windows and Linux machines. You will need a supported network card for capturing data, such as an Ethernet card or a wireless adapter. To get the latest copy of Wireshark, visit www.wireshark.org. The download page will have the proper version for your computers architecture and version operating system. A new version typically comes out every other month.

To install Wireshark, double-check the name of the file you have downloaded. If you have downloaded `Wireshark-win64-2.6.4.exe`, you will be installing Wireshark 2.6.4 for Windows 64-bit architecture. The download will include WinPcap, which allows you to capture live network traffic, not just examine saved packet captures (.pcap files).

Once you have installed the Wireshark executable, you will see the list of the different network interfaces that are functioning on the device as well as a graph to the right of current network activity on each interface. It reminds me of an electrocardiogram (EKG) that measures heart rhythms. As you see in Figure 7.1, if you have peaks and valleys, then you have traffic on that interface. If the line is flat, then that interface is not active.

When you double-click a network interface that is displaying activity, the main window will open to display all the traffic on that network. The major components of this page include the menu; the packet list, details, and bytes panes; and the status bar at the bottom, which can give you a great deal of detail regarding your capture.

The packet list pane is located in the top third of the window and by default shares information from the headers of each packet captured. Summary information includes source IP address, destination IP address, protocol in use,

length of the packet, and information about the packet. By clicking the individual packets, you control what is shown in the bottom two panes. To drill down into each packet, select the packet in the packet list pane to view more details in the middle window, which feeds data into the bottom window.

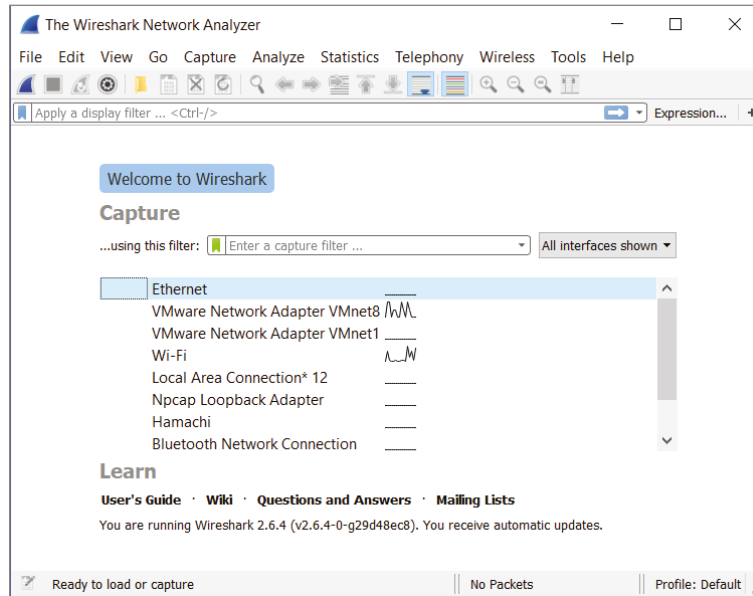


Figure 7.1: Choosing a network interface card for capture

In the packet details pane, you see individual packet size, both on the wire and bytes captured. You also see the transmission medium, protocol, source port, and destination port, and then depending on the type of packet, you may see flags or queries. You can click the > sign on the left to reveal different levels of detail about each packet in human-readable language.

At the bottom is a packet bytes pane. This displays data in hexadecimal code, which makes up the actual digital contents of the packet. It highlights the field selected above in the packet details pane. When you click any line in the middle pane, the hexadecimal code at bottom will be highlighted, giving you an extremely granular view of the data such as a URL that someone visited or contents of an email that was sent.

Under Preferences on the Edit menu, you can change the default layout of Wireshark, choosing exactly what columns you want listed; the fonts, colors, and position/direction of the panes; and what is displayed in each column. Since I learned how to use Wireshark in the default configuration, other than

making the font larger and the colors more contrasting, I usually leave all of these preferences alone.

There are also quite a few keyboard navigation shortcuts. Table 7.1 describes the common ones.

Table 7.1: Keyboard shortcuts for Wireshark

KEY COMBINATION	DESCRIPTION
Tab	Moves between packet panes
Ctrl+F8	Moves to the next packet
Ctrl+F7	Moves to the previous packet
Ctrl+.	Moves to the next packet in the same conversation (TCP, UDP)
Ctrl+,	Moves to the previous packet in the same conversation (TCP, UDP)
Backspace	In packet details, jumps to the parent node
Enter	In packet details, toggles the selected tree item
Ctrl+L	Opens capture interfaces to start a new capture
Ctrl+E	Begins a capture from Ethernet

OSI Model

The OSI model was created by the International Organization for Standardization (ISO) to give architects, engineers, and manufacturers a modular way to troubleshoot issues. Certain protocols work at certain layers of OSI. As illustrated in Figure 7.2, the OSI moves in both directions depending on whether someone is either sending or receiving data.

When data is sent across a network, the information is encapsulated as it travels down the OSI layers. When the data is received, it travels up the seven layers and is demultiplexed and delivered to the end user at the upper layers. This process is often likened to using the post office. You write a letter, fold it and put it in an envelope, address it with a destination and receiving address, pay postage, and drop it off at the post office. The post office delivers it to its destination address and the intended person.

Complex problems can be more easily solved when you take this huge process and break it into smaller pieces. Nontechnical end users will turn on their system, log in, open a browser, type in a URL, and enter a username and password to read and compose their email with no clue how it works or what it looks like from a digital point of view. For any type of analysis, it's important to understand what is happening at the different layers of the OSI model. Wireshark will capture and filter traffic on specific fields within supported protocols in manageable-sized `.pcap` files in real time.

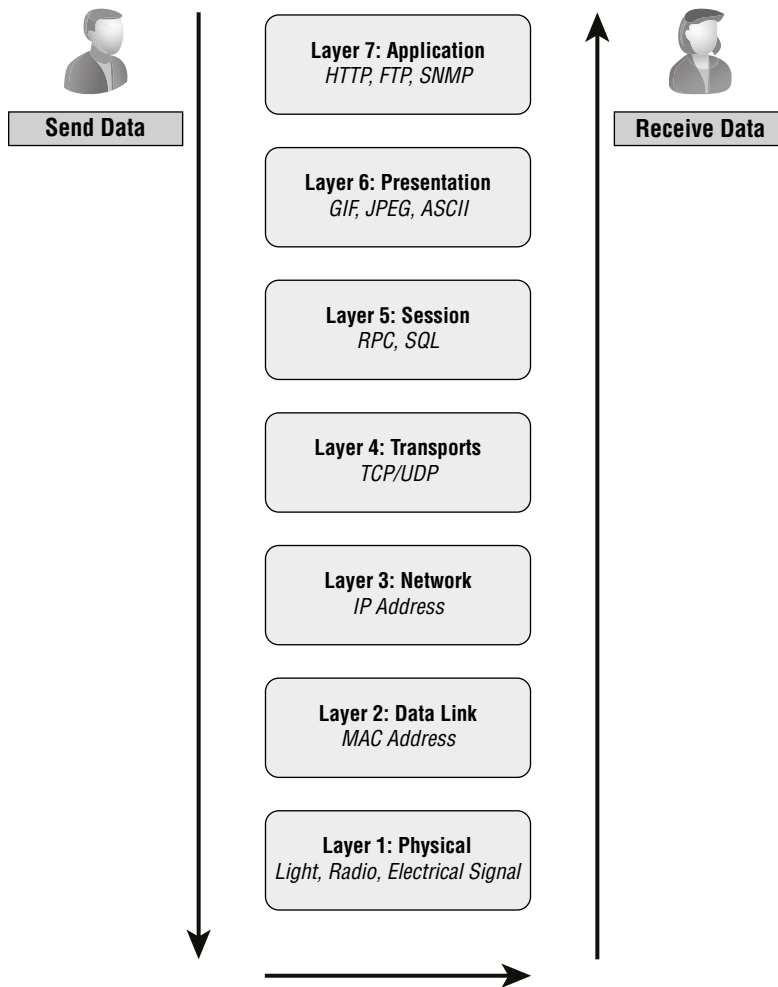


Figure 7.2: The OSI model sending and receiving data

The physical layer is where you start. This is where the transmission of data using electrical signals, light, or radio occurs. Typically you can think of this as being the hardware layer. Devices such as hubs, the actual cables, and Ethernet work at this layer. When forming a hypothesis for correcting issues in your network, the physical layer is the equivalent of “Have you turned it on?” If there is no power, you have no communication, so start troubleshooting the physical layer.

The data link layer (or layer 2) is responsible for the encoding and decoding the electrical signals from the physical layer into bits and bytes and into frames. The data link layer can be subdivided into two sublayers: MAC and Logical Link Control (LLC). The MAC layer controls how a computer on the network gains access to data, and the LLC layer controls flow and error checking. Think of MAC as the MAC address that is burned into the network interface card.

The network layer is where the switching and routing take place using IP addresses. This is where the logical path gets plotted across the World Wide Web, taking the data packet to its final destination.

The transport layer is responsible for end-to-end error recovery. TCP and UDP work to get the data where it is supposed to go, but in very different ways. Using the post office analogy again, TCP is like using return receipt requested, and UDP is the marketing material that may or may not get placed in your mailbox. TCP is connection-oriented architecture where you will see SYN, SYN-ACK, and ACK. TCP's three-way handshaking technique is often referred to as "SYN, SYN-ACK, ACK" because there are three messages transmitted. SYN is synchronize, and ACK is acknowledge. You send a packet, which is the SYN, and the receiver acknowledges the receipt of said packet, which is the SYN-ACK. You acknowledge receivers' acknowledgment that they did indeed receive that packet, which is the ACK. TCP is used to make sure systems get all the pieces they need to reassemble a message. This is called a *three-way handshake*. UDP doesn't care in the least if you receive their data. Think of a video or voice stream. Nothing gets resent if the connection breaks, and nothing is ever acknowledged that it was received. Figure 7.3 shows the ACK of packets and their number so they can be rebuilt properly by the receiver.

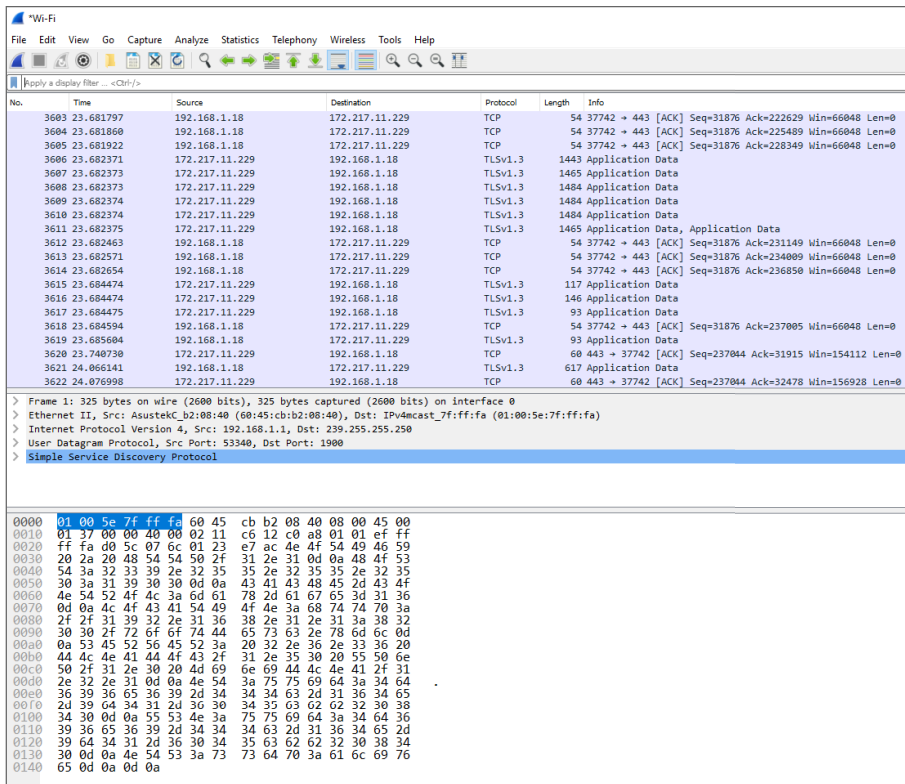


Figure 7.3: Wireshark acknowledgment traffic

The session layer is layer 5 of the OSI model. It's responsible for making, managing, and terminating connections. Layer 6 is the presentation layer, which is in charge of what gets presented to your screen. Encryption and decryption of data happen at layer 6 as well. Finally, the seventh layer is the application layer, which supports the end users and their processes. Quality of service (QoS) works at layer 7 as well as application services such as email and HTTP. QoS is the ability of a network to provide better service to certain network traffic. The primary goal is to give priority to that traffic by dedicating bandwidth to control latency.

Each layer of the OSI model ensures the delivery of data from one place to another. If a layer fails, you end up with an error. With Wireshark's help to diagnos the failing protocol, you can pinpoint where the problem is occurring so you can fix the error.

Capture

One of my favorite ways to teach Wireshark to beginners is to have students download and install Wireshark, bring up a terminal window, and capture the traffic after they launch Nmap. As you learned in Chapter 3, "Nmap: The Network Mapper," good guys as well as bad guys use it. If you can recognize what Nmap traffic looks like and you know that you're not the one running it, then odds are it is someone attempting to map out your network.

LAB 7.1: ZENMAP AND WIRESHARK

NOTE You will need to use three tools to make this lab work: a terminal window, Zenmap, and Wireshark. I am running this lab on a Windows 10 machine where I can open a command shell. You used Zenmap in Chapter 3. You can download Wireshark from www.wireshark.org.

1. Open a terminal window. Run the following command: `ipconfig /all`. Look for the IP address on your Wi-Fi network interface card.
2. Open Zenmap. In the Target field, add the IP address identified in the previous step. In the Profile field, leave the default of Intense scan.
3. Open Wireshark. On the welcome page as you saw in Figure 7.1, identify the Wi-Fi interface that corresponds with step 2. Double-click the Wi-Fi connection. It will start capturing data.
4. Go back to Zenmap and click the Scan button. On a single asset, the Nmap scan may last a to 2 minutes.
5. When the Nmap scan is done, return to Wireshark and click the red box under the word Edit. This will stop the capture, and you now have data to save and analyze.

Continues

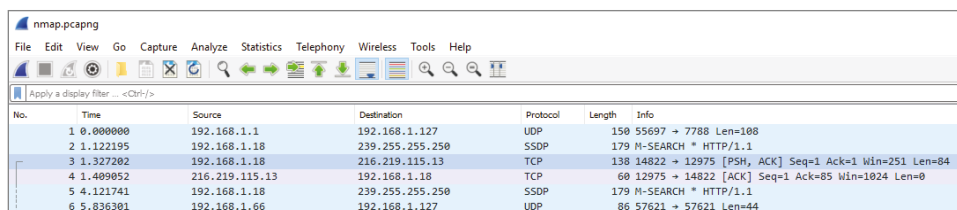
LAB 7.1 (CONTINUED)

6. With the Nmap window next to the Wireshark window, you will see traffic in Wireshark you can identify as the Nmap scan. During an Intense scan, Nmap will attempt to resolve DNS.
7. In Wireshark, look at the Protocol column for any DNS traffic. If you cannot find it by scrolling, try clicking the word *protocol* in the top pane. Each column can be sorted in ascending and then descending order just by clicking the column headings.
8. To save the network traffic you just sniffed in Wireshark, go to File ⇨ Save, name the file *nmap*, and click Save.

In any Wireshark menu, items will be grayed out if the feature isn't available. You cannot save a file if you haven't captured any data. Most of the Wireshark menu has the standard File, Edit, View, and Capture options. The Analyze menu allows you to manipulate filters, enable or disable dissection of protocols, or follow a particular stream of data. The Telephony menu is my favorite for analysis of voice traffic. In the Telephony menu, you can build flow diagrams and display statistics.

Capture filters are set before starting a packet capture. Display filters are not. In the Welcome To Wireshark window, you can find the capture filter just above the interfaces list. For instance, if you want to capture traffic only from a specific IP address, the filter would look like this: host 192.168.1.0. To capture traffic over a specific port, the filter would look like this: port 53. Double-click an interface to begin the capture.

Now that you have your first capture started, the top pane is the packet list. The first column shows relationships between packets. Figure 7.4 shows the relationships between the selected packet and other “conversations” you captured. In line 3 under the No. column, you see the first packet of a conversation represented by a right angle, and line 4 continues with a solid line. Lines 5 and 6 start with a dotted line, which signifies that these two captured packets are not part of the conversation started in lines 3 and 4.



No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.1.1	192.168.1.127	UDP	150	55097 → 7788 Len=108
2	1.122195	192.168.1.18	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
3	1.327202	192.168.1.18	216.219.115.13	TCP	138	14822 → 12975 [PSH, ACK] Seq=1 Ack=1 Win=251 Len=84
4	1.409052	216.219.115.13	192.168.1.18	TCP	60	12975 → 14822 [ACK] Seq=1 Ack=85 Win=1024 Len=0
5	4.121741	192.168.1.18	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
6	5.036301	192.168.1.66	192.168.1.127	UDP	86	57621 → 57621 Len=44

Figure 7.4: Showing conversation relationships

The next pane under the packet traffic is the packet details pane. This pane shows the protocols and fields of the packet selected in the pane above. The protocols and fields can be expanded and collapsed as needed. As you see in Figure 7.5, you can also right-click a packet for options in the packet list pane. Some fields have special generated fields such as additional information that isn't presented in the captured data, which is shown in square brackets. There will be links between packets if a relationship is found. These will be blue and underlined, and you can move from packet to packet.

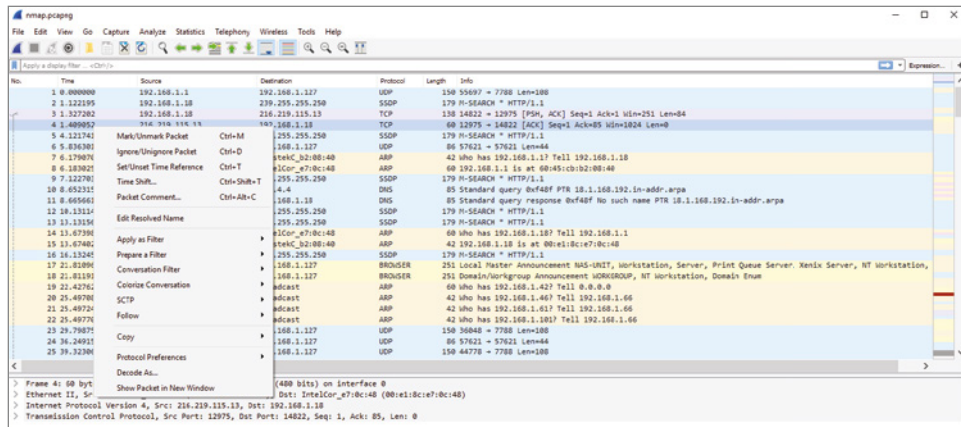


Figure 7.5: Right-clicking a packet

The packets bytes pane at the bottom of the window contains all the hexadecimal code of each packet. Each line of text contains 16 bytes. Each byte (8 bits) of packet capture is represented as a two-digit hexadecimal. In Figure 7.6, you can see the direct relationship between the IP type and the hexadecimal code.

For your second capture, repeat the steps in the preceding lab but instead of doing an Nmap scan, open the browser of your choice and navigate to `www.example.com`. The Nmap capture was slow compared to this. The second you open the browser, you see an explosion of packets as your home page loads. Navigate to another site that you usually log into, like an email account or a bank. Log in as you usually do, but watch your Wireshark traffic as you complete that task.

Since I have explained how to take a capture, it is important for me to discuss where to take a capture. If you are in a large enterprise environment and there was an issue with network performance, the placement of the network sniffer is important. Place Wireshark as close to the employees and/or customers to identify any traffic issues from their perspectives. If people are complaining about a certain server on the network, you can move Wireshark in proximity to that server to find the problem. One best practice is to put Wireshark on a laptop and move around your location while you're tracking down these problems.

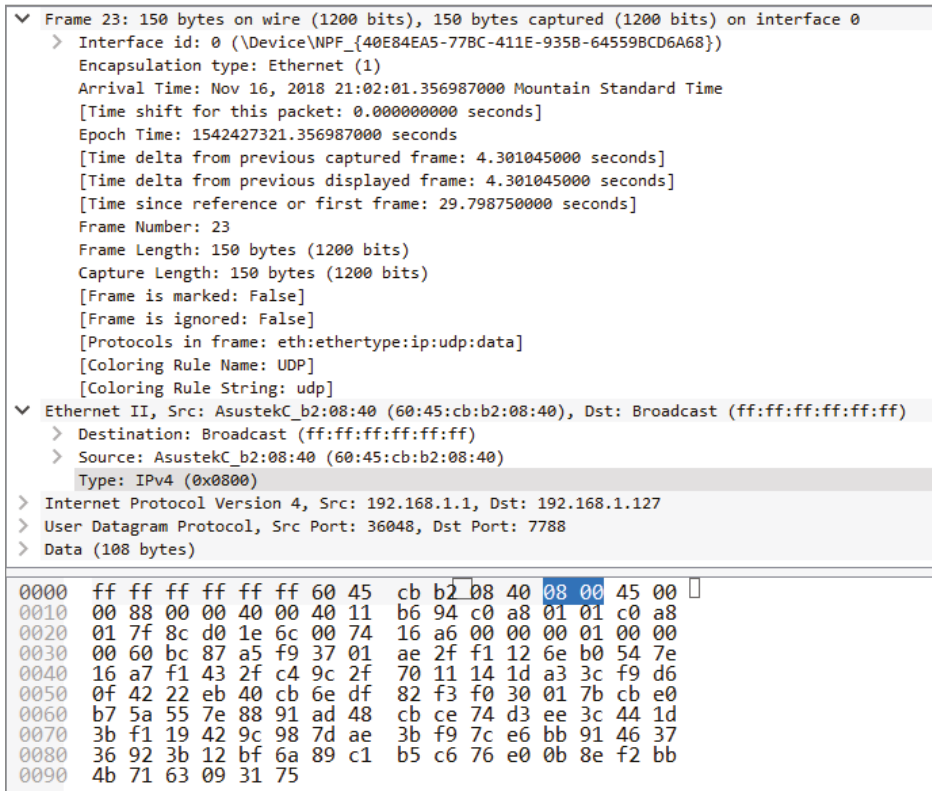
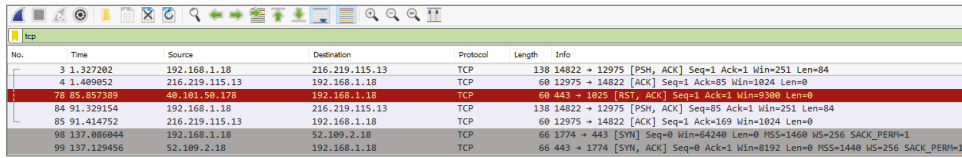


Figure 7.6: Hexadecimal representation

Filters and Colors

Wireshark uses display filters to concentrate on interesting packets while hiding the boring ones. You can select packets based on protocol, value, or comparison. To filter packets based on protocol, type in the protocol you want to narrow down to, as shown in Figure 7.7. Press Enter to accept the filter selection. When you're using a filter, it only changes the view, not the contents. The capture file remains intact. To remove a filter, click the clear button, which is the X to the right of the filter.

You can compare the values inside packets as well as combine expressions into far more specific expressions. Every field inside a packet can be used as a string, such as `tcp`. A `tcp` string will show all packets containing the TCP protocol. Once you have chosen the strings you want to knit together, you choose the appropriate operator. Table 7.2 lists commonly used filters.



The image shows a Wireshark packet capture window. The top toolbar includes icons for file operations, network analysis, and search. Below the toolbar, a list of packets is displayed, sorted by TCP traffic. The columns are: No., Time, Source, Destination, Protocol, Length, and Info. The packets are numbered 3 through 99. The 'Info' column shows details for each packet, such as '138 14822 → 12975 [PSH, ACK] Seq=1 Ack=1 Win=251 Len=84'. The packets are color-coded: red for TCP, blue for IPv4, and green for Ethernet.

No.	Time	Source	Destination	Protocol	Length	Info
3	1.327202	192.168.1.18	216.219.115.13	TCP	138	14822 → 12975 [PSH, ACK] Seq=1 Ack=1 Win=251 Len=84
4	1.489052	216.219.115.13	192.168.1.18	TCP	60	12975 → 14822 [ACK] Seq=1 Ack=85 Win=1024 Len=0
7	1.592100	192.168.1.18	216.219.115.13	TCP	60	14822 → 12975 [ACK] Seq=1 Ack=85 Win=1024 Len=0
84	91.329154	192.168.1.18	216.219.115.13	TCP	138	14822 → 12975 [PSH, ACK] Seq=0 Ack=1 Win=251 Len=84
85	91.414752	216.219.115.13	192.168.1.18	TCP	60	12975 → 14822 [ACK] Seq=1 Ack=169 Win=1024 Len=0
98	137.086044	192.168.1.18	52.109.2.18	TCP	66	1774 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
99	137.129456	52.109.2.18	192.168.1.18	TCP	66	443 → 1774 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=1440 WS=256 SACK_PERM=1

Figure 7.7: Sorting packet capture based on TCP traffic

Table 7.2: Filter operators

ENGLISH	OPERATOR	DESCRIPTION	EXAMPLE
eq	==	Equal	ip.src==192.168.1.0
ne	!=	Not equal	Ip.src!=192.168.1.0
gt	>	Greater than	frame.len>16
lt	<	Less than	frame.len<64
match	~	Field match	http.host matches
contains		Field contains	tcp contains traffic

Colorizing the traffic can be an effective filter to locate and highlight packets you may be searching for. You can choose to color packets that indicate errors, anomalies, breaches, or evidence. Wireshark has predefined coloring rules in the Edit menu under Preferences. Your coloring rules are placed at the top of the list by default, so your rules will trump any that come after.

For temporary colors, right-click a packet, go to Colorize Conversation, and slide down the list of types of traffic. To colorize the conversation, choose the protocol and select the color you would like that conversation to be. For example, you can color all IPv4 traffic blue and all Ethernet traffic red. This color rule will stay in effect until you restart Wireshark. You can also mark packets by right-clicking them. They will be shown with a black background, regardless of coloring rules. Marking a packet is helpful while analyzing a large capture, almost like a bookmark holding your place.

If you right-click a packet, you also have the ability to create packet comments. This is an excellent way to leave information that you have discovered, document a hypothesis, or communicate with other team members about network traffic you suspect is causing an issue.

Inspection

When you start inspecting and comparing packets in a packet capture, you'll notice the second column is based on time. Most computer systems start counting at 0, and Wireshark is no different. The first column is set to a time value of 0,

and all other timestamps base their times on that first packet capture. To view statistics for a number of packets, select Statistics on the menu. The statistics vary according to protocols, address, port, streams, or conversations.

A conversation is a pair of physical or logical entities communicating. Conversations can include MAC, ARP, ICMP pings, or port numbers. To compare the conversations in the packet capture, go to the Statistics tab, and then inside that menu, go to Conversations. The default tabs across the top of the Conversation dialog box will show you the data broken down into Ethernet, IPv4, IPv6, TCP, and UDP. Each line shows the values for exactly one conversation. To add other conversation statistics, click Conversation Types in the lower-right corner. When working with a large file, sorting on the bytes transferred between hosts enables you to find the most active communication based on packets or duration of conversation. In Figure 7.8, notice the column for IPv4 conversations has been sorted to show the most active conversation between source and destinations.

Wireshark - Conversations - nmap.pcapng

Ethernet - 13IPv4 - 14IPv6 - 3TCP - 3UDP - 20

Address A	Address B	Packets	Bytes	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A
192.168.1.18	239.255.255.250	15	2685	15	2685	0	0	1.122195	86.6396	247	0
192.168.1.1	192.168.1.127	8	1200	8	1200	0	0	0.000000	85.0951	112	0
192.168.1.1	192.168.1.18	11	1173	4	480	7	693	76.736320	60.3494	63	91
192.168.1.97	192.168.1.127	2	502	2	502	0	0	21.810968	0.0009	—	—
192.168.1.66	192.168.1.127	5	430	5	430	0	0	5.836301	119.8091	28	0
192.168.1.18	224.0.0.251	4	400	4	400	0	0	76.756114	0.0019	—	—
192.168.1.18	216.219.115.13	4	396	2	276	2	120	1.327202	90.0875	24	10
192.168.1.18	192.168.1.127	3	276	3	276	0	0	114.673786	1.5014	1470	0
192.168.1.18	224.0.0.22	5	270	5	270	0	0	76.737532	0.4412	4895	0
8.8.4.4	192.168.1.18	2	170	1	85	1	85	8.652315	0.0133	50 k	50 k
52.109.2.18	192.168.1.18	2	132	1	66	1	66	137.086044	0.0434	12 k	12 k
192.168.1.18	224.0.0.252	1	75	1	75	0	0	76.756871	0.0000	—	—
192.168.1.18	192.168.1.66	1	70	0	0	1	70	82.767321	0.0000	—	—
40.101.50.178	192.168.1.18	1	60	1	60	0	0	85.857389	0.0000	—	—

☐ Name resolution☐ Limit to display filter☐ Absolute start time

Conversation Types ▾

Copy ▾Follow Stream...Graph...CloseHelp

Figure 7.8: Wireshark conversations sorted by IPv4 protocol

There is another tool in Wireshark that logs anomalies found in a capture file: the Expert Info tool. The idea behind this tool is to provide a better understanding and display of notable network behavior. Both novice and expert users can solve issues quickly rather than combing through every packet manually. Expert info, as you see in Figure 7.9, is considered a hint.

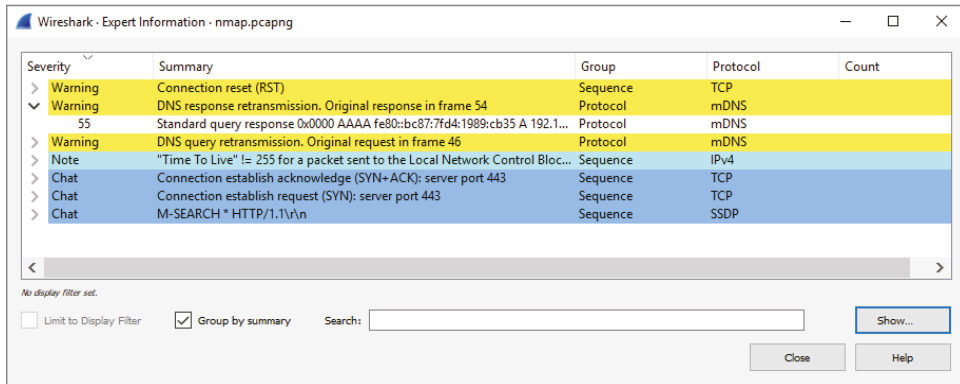


Figure 7.9: Expert Info tool color-coded “hints”

Table 7.3: Expert Info severity levels

LEVEL	COLOR	EXPLANATION
Chat	Blue	Informational, usual workflow
Note	Cyan	Normal errors
Warning	Yellow	Unusual errors
Error	Red	Serious problem

Every Expert Info type has a specific severity level. Table 7.3 lists the different Expert Info severity levels.

You can configure a graph of the captured network packets. You can configure the I/O graph to see the overall traffic as well as highs and lows in your traffic, which is typically based on a per-second, per-packet rate. You can use this to rectify problems, and you can even use it for monitoring. By default, the y-axis will set the interval to 1 second, and the y-axis will be packets like you see in Figure 7.10. Click any point on the graph to focus on that packet in the background. There are three different styles of graphs you can use: line, impulse, and dots. If you are graphing multiple items, you can choose different styles for each graph.

After capturing network traffic on your own system, the Nmap scan, and web browser traffic, if you want to branch out and look at other, more-complicated traffic but you don’t have access to a more complicated network, there is a link

inside Wireshark that will help you build a strong skill set with this tool. Under the Help menu are sample captures that can be interesting to dissect. On the page that lists the sample captures, one of the simplest to begin with is `HTTP.cap`, which is a simple HTTP request and response.

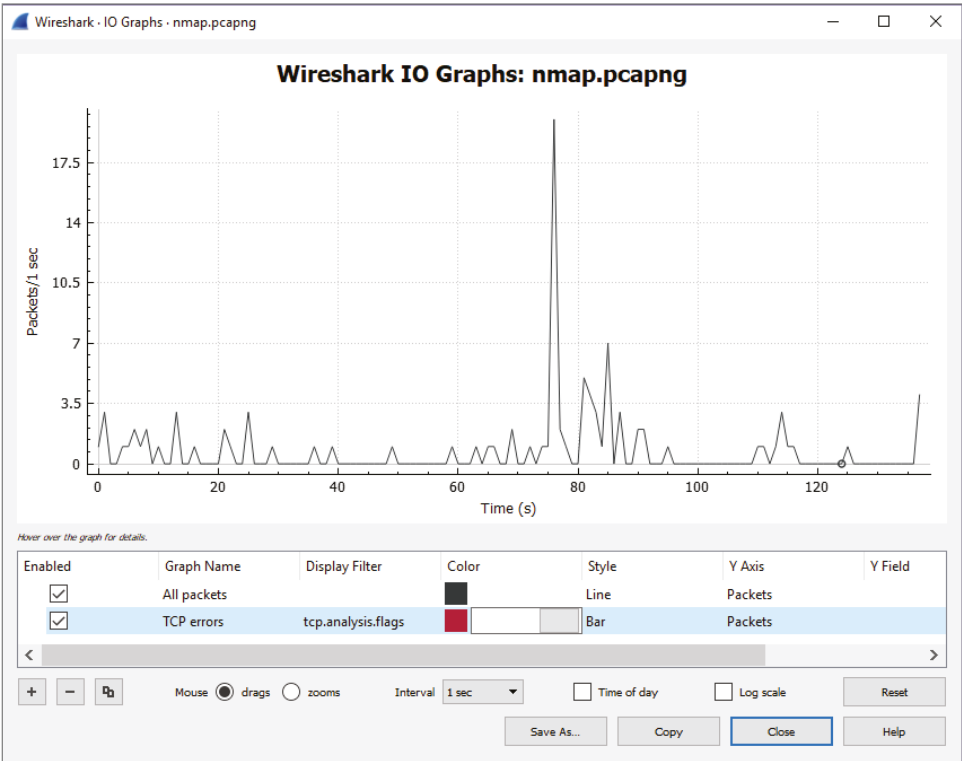


Figure 7.10: Graphing all packets versus just TCP errors